

CREDIT RISK MACHINE LEARNING

Tamweel Lite Capstone

Tamweel Lite: Credit Risk Default Prediction

Optimizing 90-Day Default Detection Under Strict Operational Review Constraints

TARGET METRIC

default_within_90d

Rare event binary target (~8% baseline rate)

OPERATIONAL CAP

12% Review Limit

Max 300 manual audit slots per batch

COST PENALTY RATIO

10 : 1 Loss Ratio

Simulated Loss = $10 \times \text{FN} + 1 \times \text{FP}$

Problem Statement

Predict financing defaults within 90 days by identifying which financing applications are likely to be successfully repaid and which are likely to default. The AI model will estimate the probability of default for each application and help identify high-risk customers for further review, while operating within a limited manual-review capacity.



Robust Methodology & Leakage Prevention



Temporal Forward Splits

Train on historical cohorts, evaluate on future horizons. Prevents temporal lookahead bias in model evaluation.

TIME-AWARE VALIDATION



90-Day Label Buffer

Mandatory outcome observation window enforces mature label collection before training cutoff.

LABEL MATURITY GUARD



Pipeline Isolation & Dedup

Scikit-Learn pipeline encapsulation for imputers and scalers; strict customer ID grouping eliminates train-test bleed.

ZERO DATA LEAKAGE

Methodology Guarantee: Prevents data leakage and ensures realistic out-of-time performance estimation.

LEAKAGE AUDITED

Model Selection & Ensemble Gate Decision

WINNING ARCHITECTURE

Single Logistic Regression

0.3917

Mean Average Precision (AP)

Outperformed gradient boosting and tree baselines in ranking precision across cross-validation folds.

GATE FAILURE

Ensembles Rejected

$r = 0.98$

Error Residual Correlation

Models made identical mistakes. Ensembles failed to yield meaningful lift ().

$\Delta AP < 0.0298$

GOVERNANCE CHOICE

KEEP SINGLE Policy

12% Cap

Operational Audit Ceiling

Eliminated operational complexity and latency while maintaining full linear interpretability.

Probability Calibration & SHAP Explainability

Platt Scaling (Sigmoid)



Transforms raw model logits into true default probabilities using cross-validated logistic calibration.

- ✓ **ECE Reduction:** Minimizes Expected Calibration Error across risk deciles
- ✓ **Reliability:** A predicted 15% risk equals 15% true defaults

CALIBRATED PROBABILITIES

SHAP Transparency



Provides game-theoretic local feature attributions for regulatory compliance and adverse action notices.

- ✓ **Beeswarm Plots:** Ranks global risk drivers across the portfolio
- ✓ **Waterfall Plots:** Breaks down individual loan risk decisions

EXPLAINABLE ML

Final Policy, Limitations & Monitoring

POLICY RULE

Calibrated Threshold ≥ 0.1223

Optimizes simulated loss () while capping FP manual reviews at strictly 300 applications.

KNOWN LIMITATION

Synthetic Data Shift

Synthetic feature distributions require real-world validation prior to full automated lending deployment.

MONITORING PLAN

PSI & Drift Audit

Track Population Stability Index (PSI) and score drift monthly with quarterly model retraining triggers.

Tamweel Lite: Conclusion

DEVELOPER

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PROJECT TYPE

**Individual Capstone
Project**

TRAINING PROGRAMME

**SDA-DSC-211 — Advanced Machine
Learning Methods**

Thank You